ChatGPT

Waste Object Characterization and Mass Estimation: Methods and Implementation

Introduction

The ability to automatically recognize waste items and estimate their volume and mass has significant practical applications in recycling, smart bins, and waste management. Recent advances in computer vision and sensor fusion have enabled systems that can **classify waste by material or object type, measure its dimensions, and even infer weight**, all in real time on embedded devices. This report surveys the **state-of-the-art algorithms, tools, datasets, and implementation strategies** for waste object characterization and mass estimation. We cover academic research and open-source projects, including **RGB vision methods, depth sensing, sensor fusion (e.g. weight scales, acoustic, infrared sensors), and multimodal techniques**. Key tasks such as **waste classification, material detection, object segmentation**, **volume estimation (2D, 3D, monocular depth)**, and **mass estimation (via regression, density models, or physics-based approaches)** are reviewed. We also discuss **notable datasets** (TrashNet, TACO, etc.), **modern depth estimation models** (e.g. MiDaS, Depth Anything), **vision-language approaches** (CLIP, BLIP, SAM), and **practical deployment considerations** on edge AI hardware (NVIDIA Jetson, Google Coral TPU, Raspberry Pi with Hailo). Finally, we outline **evaluation metrics and protocols** for benchmarking such systems. The goal is to inform the development of real-world productizable systems (smart bins, edge sorting stations) with a comprehensive understanding of current methods and best practices.

Waste Recognition Methods (Vision-Based)

Waste object recognition can be approached at multiple granularities: basic classification of an item's category or material, object detection in a scene, and pixel-wise segmentation of waste objects. Classical image classification models (CNNs) have been widely applied to waste sorting tasks, while detection and segmentation models allow locating and identifying multiple items in cluttered scenes.

Image Classification and Material Detection

Early work on automated waste sorting focused on classifying single items. For example, **TrashNet** (Thung & Yang 2016) provided a dataset of ~2,527 images in 6 classes (paper, plastic, glass, metal, cardboard, trash) for classification [1]. Traditional models using SIFT features with SVM achieved ~63% accuracy, while a basic CNN reached 87% on TrashNet [2]. Subsequent research improved on these results using deep transfer learning. Adedeji & Wang (2018) applied a pre-trained ResNet-50 CNN, achieving ~87% on TrashNet as well [3]. More recent studies report even higher accuracy by optimizing architectures or using meta-heuristic tuning – e.g. **DenseNet-121 with a genetic algorithm** reached 99.4% on TrashNet [4], and a hybrid model (DSCam) achieved 98.9% [5] [6]. These high results, however, often come from training on the limited classes in lab settings and may not generalize to diverse real waste streams [7].

Material-focused classification is also explored. Instead of specific product categories, models classify the material type (plastic, paper, metal, organic, etc.). For instance, **Zhang et al. (2021)** designed a model to sort household waste into four broad categories (Recyclable, Residual, Organic, Hazardous), achieving ~94.7% sorting accuracy in a smart bin prototype [8] [9]. In some cases, non-visual sensors aid material identification: **infrared spectroscopy** has even been coupled with CNNs (PlasticNet) to classify plastic types with near 100% accuracy in controlled settings [10], though such specialized sensors (ATR-FTIR) are less common in consumer bins. Overall, **transfer learning** on large CNNs (ResNet, EfficientNet, etc.) and fine-tuning have been the dominant approach for image-based waste classification, often yielding 90–99% accuracy on benchmark datasets [11] [12]. However, achieving robust performance in real-world conditions requires addressing background clutter, occlusion, and a long-tail of unseen items – challenges that single-image classifiers struggle with if limited to a few classes [7].

Object Detection for Waste Localization

Object detection models identify and locate waste items in an image, outputting bounding boxes and class labels. This is crucial for scenes with multiple pieces of litter or waste on conveyor belts. Modern detectors like **YOLO, Faster R-CNN, and SSD** have been applied to waste detection. For example, **Mask R-CNN** was used in the TACO dataset toolkit to detect and segment litter in the wild [13]. **YOLOv2** (one-stage detector) was successfully used by Sterkens et al. to detect electronic waste (batteries) in X-ray scans with 91% true-positive rate [14]. For RGB images, researchers have trained YOLO and Faster R-CNN on custom waste datasets. Saeed et al. (2020) combined Faster R-CNN with edge computing for detecting street litter in real time [15]. Lin et al. (2022) proposed a specialized detector (DSCam) for trash, outperforming baseline CNNs [6]. A challenge in waste detection is the variability in object size, shape, and overlap. One strategy is **ensemble or cascaded models**: Li et al. (2022) introduced a Multi-Model Cascaded CNN (MCCNN) that fuses YOLOv4, Faster R-CNN, and DSSDs sequentially to handle different scales [16] [17]. They curated a large-scale waste image dataset (LSWID, 30,000 images, 52 categories) and reported ~10% higher detection precision by combining detectors [18] [19]. Such cascades exploit different models' strengths (e.g. one better at small objects, one at large) to improve overall detection in complex scenes [20] [17].

Instance segmentation goes further by providing a mask for each object's silhouette. The **TACO (Trash Annotations in Context)** dataset is a prominent benchmark here, containing 1,500 images of real litter in outdoor scenes with ~5,000 annotated instances spanning 60 categories [21] [22]. TACO's masks have been used to train segmentation models that can distinguish, for example, a plastic bottle versus a can even if they are partially deformed [23]. **Mask R-CNN** with a ResNet-50 backbone has been applied to TACO, and the dataset's authors provide a toolkit to train and evaluate detection+segmentation on this data [13]. Figure 1 shows an example of segmented litter from TACO (a plastic bag floating in water). Multi-object segmentation allows systems to count items and measure visible areas, which can aid downstream volume or material estimation.

Figure 1: Example of instance segmentation of litter from the TACO dataset (green mask on a "Single-use carrier bag"). High-resolution annotations enable training models to detect and segment waste objects in context, despite cluttered backgrounds.

Advanced techniques are emerging to improve open-world detection of waste. The **Segment Anything Model (SAM)**, a recent foundation model for segmentation, can generate masks for any objects without training. By combining SAM with a classifier (or CLIP, discussed below), one can detect arbitrary waste items and then identify their category by matching the image patch to known waste classes. This **SAM+CLIP fusion** enables open-vocabulary detection – for example, segmenting an unknown object and then recognizing it as "a piece of cardboard" via CLIP's text-image similarity. Such approaches are promising for **flexible, adaptable waste recognition** where the system might encounter new packaging or products not in its training data. Research on **promptable segmentation and open-vocabulary detection** is ongoing, but initial work (e.g. Grounded-SAM) demonstrates that combining segmentation and vision-language models can achieve accurate detection of novel litter categories [24] [25] .

Multimodal and Vision-Language Recognition

In addition to images, other sensing modalities can enhance waste identification. **Acoustic sensors** provide one intriguing modality: the sound of an item being tossed or crushed can reveal its material. Lu et al. (2020) demonstrated that a 1D CNN processing audio signals of waste items being discarded achieved **92.4% accuracy** in classifying material, outperforming traditional MFCC+SVM baselines [26] . The distinct sound of glass shattering vs. a can thudding can thus serve as a reliable cue. Other studies have combined **visual and acoustic data** for multimodal classification, reporting accuracies ~95–96% for 12 waste types when using both camera images and microphone data [27] . **Infrared (IR) sensors** can also play a role: near-IR reflectance can differentiate materials (as used in industrial recycling for polymer identification [28]). While IR spectroscopy is expensive, simple IR distance sensors are common for detecting the presence of waste (e.g. a break-beam sensor to trigger a camera when an object falls into a bin).

A notable recent trend is applying **Vision-Language Models (VLMs)** to waste recognition. Models like **OpenAI's CLIP** are pre-trained on hundreds of millions of image-text pairs and can classify images in a **zero-shot** manner using text prompts [29] [30] . This capability is highly relevant for waste sorting, which may involve an open set of categories. Laurençon et al. (2024) leveraged CLIP and similar VLMs for general waste classification, crafting prompts such as “a photo of a plastic bottle” or “a piece of torn paper” to guide the model [31] [32]. Without any task-specific training, CLIP reached about **72–84% accuracy** on a 5-class waste test set [33] [34]. By optimizing the text prompts (prompt engineering), the zero-shot accuracy was further boosted to **90.5%** on the best model [35] [36]. This impressive result highlights that **VLMs can generalize to waste categories using semantic cues** – e.g. knowing what “cardboard” or “plastic bag” typically look like from web data [37] [38]. Vision-language models thus enable scalable recognition without collecting extensive labeled waste images for each new category. In addition, models like **BLIP** (which can generate captions or answer questions about images) could describe an unknown waste item in natural language, aiding material reasoning (e.g. noting if an object “looks like metal”). While these VLM techniques are cutting-edge, they are increasingly feasible to deploy on the edge via distilled or optimized versions (OpenCLIP, etc.), and they offer a powerful way to handle the long tail of waste object diversity [29] [30].

Volume and Dimension Estimation Techniques

Estimating the **volume or physical dimensions** of waste objects is important for assessing container fill levels and for weight inference. Several approaches exist, depending on available sensors:

- **2D image-based estimation:** Using a single RGB image, one can attempt to infer object size through known references or learning-based methods. For example, if the camera is calibrated and the object is on a known background, simple geometry (pixel size vs known reference size) can estimate dimensions. However, without depth information or a reference object, 2D alone gives only relative size. Researchers have tried using object detection bounding box size as a proxy for volume, but this is unreliable due to perspective and unknown scale. Some systems allow a user to input distance or select keypoints on the image for measurement. A recent approach by Wimalasiri et al. (2024) used **MobileNetV3 and MiDaS depth** to let users click two points on an object in an image, then computed the distance between those points by combining depth map values and pixel coordinates. Such interactive measurement can yield real-world lengths without a physical ruler, but it’s semi-automated.

- **Stereo vision:** Using two cameras (or a stereo module) can provide depth via triangulation. Stereo vision has been used for volume estimation in scenarios like piled waste or food portions. It requires calibrating the cameras and computing a disparity map. For instance, research on food portion size sometimes uses stereo rigs to estimate volume of food on a plate [39]. In waste management, **stereo cameras** could map the shape of an item or a heap of trash; one study estimated lettuce weights using a 3D stereoscopic technique, extracting volume features to predict weight [40]. Stereo is **more hardware-complex** than monocular but can yield actual metric depth if calibrated. It struggles with textureless surfaces or low lighting, but for well-textured waste items (printed packaging, etc.) it can produce dense 3D point clouds.

- **Depth sensors (RGB-D and LiDAR):** Active depth sensors like **Intel RealSense** cameras (which use stereo IR or time-of-flight) or Microsoft Kinect can directly capture a depth map of the scene. This greatly simplifies volume estimation, as the dimensions of objects can be measured in 3D by analyzing the depth data. A straightforward method is to segment the object in the depth image and **integrate the depth over its area**. For example, Kim et al. (2025) propose a method to compute volume from a single RGB-D frame by summing small voxel volumes across the segmented object region [41] [42]. They first use a pre-trained segmentation model on the color image to isolate the object, then take the depth values for those pixels and subtract the background plane. The object’s volume is obtained by dividing the region into 2×2 pixel blocks and feeding the heights in each block into a small neural network that predicts the volume of that column [43] [42]. Summing these gives the total volume, with an average error of only ~2.37% in their tests [44] [45]. This demonstrates that **RGB-D sensors can achieve highly accurate volume measurements** for irregular objects, essentially by treating the depth map as a stack of tiny pillars and learning a correction for each pillar's volume. In industrial settings, **3D LiDAR scanners** or structured-light sensors can capture waste pile shapes for volume. They are more expensive but offer high accuracy even for large volumes (e.g. measuring a heap of material in a dumpster). For instance, solid-state LiDAR has been used to measure volumes of bulk materials like soil with errors under a few percent [46], and similar principles can apply to waste.

- **Monocular depth estimation:** When a physical depth sensor is not available, **AI-based monocular depth estimation** can infer approximate depth from a single RGB image. Models like **MiDaS** (Microsoft's model for monocular depth) and the newer **Depth Anything** model have been trained on large datasets to predict a dense depth map from an image. MiDaS provides **relative depth** – a depth map up to scale – which is useful for understanding object shape but not directly for measuring volume in absolute units. **Depth Anything** (2023) is a foundation model trained on 1.5 million labeled depth images and 62 million unlabeled images. It offers both relative depth and a **metric depth mode** that has been fine-tuned on datasets like NYU-Depth-V2 and KITTI to produce scaled depth outputs. In practice, using monocular depth for volume estimation still requires calibration: the output scale might not be perfect, especially if the camera's field of view differs from training data. Nonetheless, Depth Anything and MiDaS can be used as a plug-in to estimate distances and shape. For example, in an edge-based waste measurement project, Depth Anything was used to generate point clouds of waste items from a Jetson Nano camera feed [47] [48]. By calibrating the camera (e.g. known chessboard, etc.) and possibly fine-tuning the depth model in the specific setting, one can attain approximate real-world dimensions from monocular images. A noted issue is that **monocular models output depth with some arbitrary scale or bias**, so additional steps (like known object size in view or sensor fusion) may be needed to get absolute measurements.

- **Ultrasonic and fill-level sensors:** In smart bins and large storage bunkers, **ultrasonic ranging sensors** are commonly used to estimate volume of waste by measuring the fill level (distance to the top of the pile) [49] [50]. They are low-cost and easy to deploy (e.g. multiple ultrasonic sensors mounted on a bin lid or above a bunker). Figure 2 illustrates an example where an array of ultrasonic sensors measures the profile of a waste pile in a sorting facility. By scanning at multiple points, the system can get a rough volume estimate of irregular piles [51] [52]. However, accuracy is often limited: irregular shapes, varying material (sound absorption), and movement cause errors [53] [54]. Still, for **bin fill-level monitoring** (e.g. alert when a bin is ~80% full), a single ultrasonic sensor reading distance to the trash surface can be a pragmatic solution. New optical Time-of-Flight (ToF) sensors or even simple IR proximity sensors can similarly gauge a bin's fill percentage.

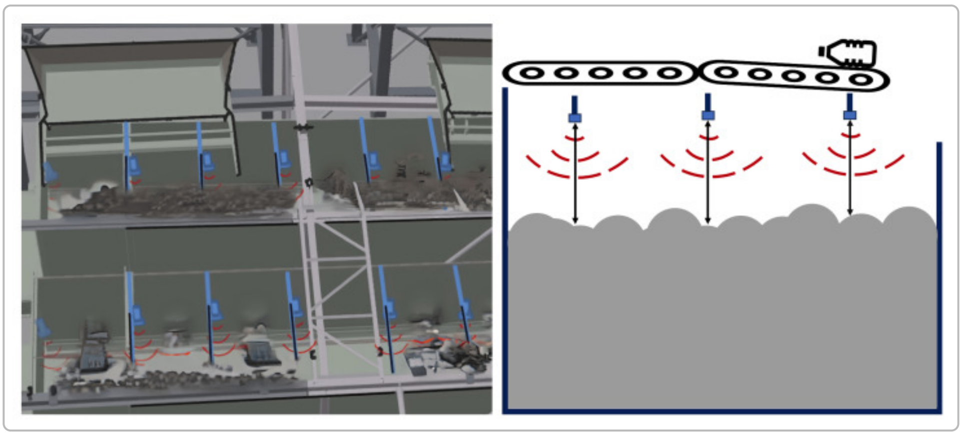

Figure 2: Volume estimation in waste sorting using range sensors [49] [52] *. Multiple ultrasonic sensors (blue boxes) mounted above a bunker measure distances (red arcs), providing a cross-sectional profile of the waste pile. Such arrays help estimate bulk volume, though irregular shapes and sensor noise introduce challenges.*

In summary, **depth information (whether via stereo, depth camera, or learned monocular models) is extremely valuable for measuring waste dimensions and volume**. For single items, an accurate depth map allows computing the object's 3D bounding box or mesh and thus its volume (assuming known calibration). For piles of loose waste, multiple depth readings across the surface can be integrated to estimate total volume. Each method balances cost and precision: monocular depth is cheapest but only approximate, stereo and structured light give metric reconstruction with moderate cost, and LiDAR gives high precision at high cost. The choice depends on the application – e.g. a consumer smart bin might use a tiny stereo camera or monocular depth AI to keep costs down, whereas an industrial sorter might use LiDAR or multiple cameras for accuracy.

Mass Estimation Approaches

Estimating the **mass (weight)** of waste objects is a complex task, as it depends on both the object's volume and its material density. Several strategies have been proposed, ranging from direct prediction by AI to physics-based calculations and sensor fusion.

Vision-Based Mass Regression (Image-to-Weight)

One line of research attempts to predict an object's mass directly from an image, treating it as a regression problem often called **"image2mass."** Standley et al. (2017) pioneered this by creating a dataset of everyday objects with known dimensions and weights (sourced from Amazon product listings) [55] [56] . They extracted 14 hand-crafted features from each object's image (likely relating to size and color) and used **two Xception CNNs** to estimate the object's volume and density, which were multiplied to predict mass [57] [58] . Their system achieved a coefficient of determination R² ≈ 0.69 (and ~67.5% minimum ratio error) on novel objects [59] [60] , outperforming untrained human guesses [57] . This **image2mass approach** demonstrated that vision alone can capture some cues for weight (larger objects or denser-looking materials weigh more), but also highlighted its limits – the model was far from perfectly accurate. Subsequent work has extended image2mass with additional inputs. For example, **Santley et al. (2017)** reported using two CNNs: one predicting object volume from the image, the other predicting material type (hence density), combining them for mass [58] . Adding material information improved accuracy, as expected.

In domains like agriculture and food, image-based weight estimation has seen promising results. **Utasi et al. (2019)** fed image features (area, dimensions) into an ANN to predict weights of fruits, obtaining

R² of 97–99% for certain produce [61]. In these cases, objects (fruits) were more uniform in shape and the imaging setup provided consistent scaling, enabling high accuracy. In the waste context, items are more variable, so pure image regression typically has higher error. Still, the approach is viable when other sensors are unavailable. A recent example in food: Wimalasiri et al. (2024) combined **Faster R-CNN for object detection and MobileNetV3 for weight regression** to estimate the weight of food portions from a single image [62] [63]. On a dataset of 14 food types with varying portion sizes, their model achieved a mean absolute percentage error of 0.064% and R² = 98.65% [64] [65] – essentially very precise, thanks to controlled conditions and the model learning the visual appearance of quantity. This "two-stage" approach (detect item then predict weight) could be applied to waste: e.g., detect a bottle in the image, then use a specialized network to infer if it's full or empty, large or small, which correlates with weight. Indeed, **multi-output CNNs** that jointly classify an object and estimate a continuous value have been explored for recycling. Diaz-Romero et al. (2021) developed a **multi-output DenseNet** to simultaneously classify scrap metal type and predict its mass, using combined RGB and 3D inputs [66] [67]. Such multi-task learning can share visual features between identification and weight estimation, potentially improving both. In their case, fusing RGB and depth data improved classification to 98% and provided a basis for mass estimation [68].

In summary, **direct image-to-weight models** (often CNN regressors) are feasible when trained on specific datasets, but their accuracy depends heavily on having representative training data for the items whose weight will be predicted. They also function as "black boxes," which can be hard to generalize outside their training distribution (e.g., a network trained on images of filled vs empty bottles might misestimate a half-full bottle if not seen before). Therefore, many systems incorporate more structured reasoning or additional sensors for mass estimation.

Volume & Density Models (Physics-Based Reasoning)

A more explainable approach is to compute weight as **mass = volume × density**, which requires estimating the object's volume and identifying its material (to look up an appropriate density). This approach was suggested in early image2mass work: if one can get the 3D volume from vision and know what material the object is (plastic, glass, metal, etc.), mass can be estimated [69] [70]. For example, a crumpled aluminum can and a plastic bottle of the same size have different densities (aluminum ~2.7 g/cc vs PET plastic ~1.3 g/cc), so material classification is crucial. In practice, this method faces challenges: waste items are often **irregularly shaped or hollow**, so computing their exact volume is non-trivial. If the object is a bag or container with air inside, volume times density would overestimate the weight (since much of the volume is empty). Standley et al. (2017) encountered this issue – their method effectively tried to infer an "effective density" of objects to account for hollowness [71] [72]. Still, in controlled scenarios, volume×density can work well. For instance, Konovalov et al. (2019) segmented fish in images and computed the fish area and thickness to estimate volume, then used known fish tissue density to get weight, achieving ~4.36% error [73] [74]. For waste, one could imagine: measure an object's dimensions via depth, assume a simple shape (e.g. treat a plastic bottle as a cylinder) to approximate volume, then multiply by a typical density for plastic. This would yield a rough weight estimate. The accuracy of density-based models hinges on correctly identifying material and assuming a reasonable solid fill. For solid metal parts or glass, it could be quite accurate; for containers or crumpled items, one might incorporate a factor for hollowness (perhaps via machine learning calibration).

In research, **hybrid models** combine learned and physics steps. Zhang et al. (2020) did this for fish: they extracted features from images, used PCA to reduce them, and then a backpropagation neural network (BPNN) to predict fish weight, effectively blending analytical features (like dimensions) with a learned model [75] [76]. They achieved MAE ~0.01 (relative to fish weight of ~0.5 kg, so ~2% error) [77]. A similar approach for waste might involve features such as detected object size, predicted material class, maybe estimated volume from depth, fed into a regression model (SVR, BPNN, etc.) to output weight. This was exemplified by Diaz-Romero et al. (2021) for scrap metal: they used 24 features extracted from 3D images (like bounding box size, surface area, etc.) and combined a CNN (for visual features) with a BPNN that learned to predict mass [78] [79]. Their best model achieved R² ~0.71 on mixed metal scrap [80]. This is a moderate correlation, reflecting the difficulty of mass prediction for arbitrary shapes – but it was significantly better than using density*volume alone without learning [70] [72].

In practical smart bins, a simplified volume×density approach can be implemented if the system can at least **categorize the item by material** (via vision or other sensors). For example, once an object is classified as "plastic bottle" vs "glass bottle," the system could assign a nominal density (plastic ~0.9 g/cc if crushed, glass ~2.5 g/cc) and multiply by estimated volume (from dimensions or known average volume of such bottles). Some bins might even maintain an *internal database*: e.g., if the vision system recognizes a specific product (barcode or shape), the system could retrieve its known weight (some recycling bins do this for common beverage containers). This moves towards an **identification-based approach** rather than purely inferring weight from appearance.

Sensor Fusion with Weight Sensors

The most direct way to get mass is, of course, to **measure it with a scale**. Many productizable systems incorporate a load cell or force sensor in the bin. For example, a **smart bin (TrashBot)** may have a weighing platform that records the weight change when an item is deposited [81] [82]. By fusing this with the vision system, one can achieve very accurate mass estimation. The vision can identify each item and the scale gives precise weight; together, the system can log "bottle – 0.5 kg", "cup – 30 g", etc. This data could even refine the model's internal density assumptions over time. **Edge inference stations** in recycling centers often use this: an object passes under a camera and then onto a scale or through a weight-sensitive chute. The downside is added cost and complexity (calibration of the scale, dealing with multiple items at once, etc.). Some designs use a **"weigh as you throw"** method – the bin weighs itself before and after an item is dropped, the difference is the item's weight. This requires ensuring only one item is added at a time (which smart bins often enforce by intake mechanism).

Beyond static weighing, more exotic sensor fusion has been tested: **impact-based estimation** where an accelerometer measures the force of impact when the trash hits a surface, correlating with weight; or **acoustic signatures** where heavier objects might produce lower-frequency thuds. These are not yet common, but research like **Tongaraas (2020)** used tongs with a pressure sensor and a microphone to identify litter type by the sound and resistance when the item is grasped [83]. While not exactly mass estimation, it shows how combining tactile and auditory data can infer properties of waste.

In summary, **sensor fusion approaches** often yield the best mass estimates: using vision to classify the item and a scale to measure weight directly is the gold standard. If a scale is not feasible, combining vision-estimated volume with another cue (like acoustic or an IMU for heft) can reduce uncertainty. Importantly, if the vision system has already categorized the object, the mass estimation problem space narrows – e.g., if it knows the item is a "plastic water bottle of 1L size", it can estimate whether it's full or empty by color or sound, etc., and use a typical weight for that item.

Datasets and Tools for Model Development

Research and development in this area have benefited from several **open datasets** and tools. We've mentioned **TrashNet** (classification) and **TACO** (detection/segmentation). A few others are noteworthy:

- **AquaTrash:** A smaller dataset (369 images) of aquatic waste (marine debris) in 4 categories [19], aimed at litter in water environments. It's used to test models' ability to generalize to different backgrounds (water, beaches).

- **AutoTrash dataset:** Created by Donovan (2016) as part of an automated sorting trash can project [84]. It had around 100 photos in each of 50 categories, focusing on common recyclables, but primarily used for a Pi-based classifier distinguishing compost vs recycling [85]. It's not widely available but sparked interest in *DIY smart bin* implementations.

- **VN-Trash:** A dataset mentioned in literature [19], likely a Vietnam-sourced waste image set. Details are scarce in open literature, but it underscores the need for region-specific data (as waste appearance can vary by locale).

- **Synthetic Datasets & Augmentation:** Due to limited real data, synthetic data generation is popular. One approach is **cut-and-paste augmentation**: *WasteGAN* (2020) is a method that segments waste objects from one set of images and randomly pastes them on new backgrounds to create diverse training images [86] [87]. Similarly, **BackRep** (MDPI 2021) took images of waste on a white background and replaced the background with random scenes to mimic context [88]. These techniques effectively multiply the training data and help models handle varied environments. Some works also use **GANs to generate waste images** that resemble the distribution of real trash photos [89]. Synthetic depth data can be generated via simulation as well – e.g., rendering 3D models of bottles and boxes to train depth estimation or detection models.

- **image2mass dataset:** As mentioned, Santley et al.'s dataset of Amazon products is one of the few with both **image and ground-truth weight** for each object [55] [59]. While not publicly released, it consisted of 1476 images of 479 household items [57]. This idea has been extended by others – for instance, a recent work added **material property embeddings** to the image2mass model to improve accuracy [90], implying they had to gather or label material info for each item. For waste-specific image-to-weight, there is still a gap: no large public dataset provides paired images and weights of trash items. This is why many projects create a **custom measurement dataset**, as in the Jetson Nano project outline which suggests collecting household waste items, recording their images, dimensions, and weight for training a regression model [91] [92]. Even a few hundred samples can help calibrate a vision model for mass estimation.

- **Roboflow Universe** and others: In the open-source community, repositories like Roboflow Universe host several user-contributed waste datasets (e.g. "Trash Detection" with ~2.7k images, "Ocean Waste" with 9k images) covering various waste types [93] [94]. Pre-trained models (YOLOv5/YOLOv8, etc.) fine-tuned on these are sometimes shared, which can be a starting point for developers. The community recommends using new tools like **Autodistill with Grounded-SAM** (which leverages SAM+CLIP) to **bootstrap object detectors** for waste without extensive labeling [95].

Tools and frameworks: Implementing these systems involves combining multiple components. Common tools include: - **Deep learning frameworks**: PyTorch and TensorFlow/Keras are heavily used for model training. Many academic papers prototype in these. For deployment on edge, TensorRT (NVIDIA) or TFLite (TensorFlow Lite) is used to optimize models. - **Pre-trained models**: Models like CLIP, MiDaS, Depth Anything, YOLO (various versions) are available open-source. The **MiDaS** GitHub provides code to compute depth from a single image easily [96], and **Depth-Anything** has released weights (v2 supports even 4K images with prompting) [97] [98]. - **Hardware integration**: Using an Intel RealSense camera, for instance, can be done via the RealSense SDK to get RGB and depth frames. Load cells interface via microcontroller (Arduino or Raspberry Pi ADC) to feed weight data. Projects often use **OpenCV** for image processing (e.g. for calibration, edge detection in measurement tasks). - **Cloud-based auto ML**: Some city deployments have tried cloud vision APIs for waste classification (e.g., Google AutoML Vision trained on trash images). However, due to latency and privacy, the trend is toward on-device inference.

Edge Deployment Considerations

For real-world deployments like smart recycling bins or sorting robots, running the AI algorithms on **edge hardware** is crucial. These environments have constraints on power, cost, and compute, and require the system to be **real-time and reliable**.

Embedded AI hardware options: - **NVIDIA Jetson family:** The Jetson Nano (128 CUDA cores, 4 GB RAM) is a popular low-cost choice for prototypes, capable of running lightweight CNNs at a few frames per second. More powerful models like Jetson Xavier NX or Jetson Orin (with up to 1024 CUDA cores and Tensor Cores) can handle heavier models and even multiple networks in parallel. For instance, a Jetson Orin Nano can run Depth estimation + Object Detection concurrently, but one must choose model sizes carefully: one benchmark found that a YOLOv8-medium model was too slow on the Orin Nano, whereas YOLOv8-n or YOLOv8-s (smaller variants) ran acceptably [99]. Thus, **model compression and selection** is key – e.g., using a smaller backbone (MobileNet, EfficientNet-lite) or fewer channels to meet timing. Jetson devices support TensorRT optimization, which can give 2-4× speedups by FP16 or INT8 optimization of the models.

- **Google Coral Edge TPU:** A USB or PCIe accelerator that can run quantized TensorFlow Lite models at very low power. Coral TPUs excel at image classification and detection models that have been quantized to INT8. For example, a MobileNet or EfficientDet can run at tens of FPS on a Coral. The challenge is the model must be converted and fit in the Edge TPU memory (~8 MB per model). This could be suitable for a simpler waste classifier (Edge TPU could classify an item's image in a fraction of a second). Coral is often paired with a Raspberry Pi for a compact setup.

- **Hailo-8 AI module:** An up-and-coming accelerator that can be used with Raspberry Pi or other hosts, designed for running multiple neural networks simultaneously with high efficiency. A Hailo-8 can deliver up to 26 TOPS (trillion ops/sec) in a few-watt module, which might allow running detection, depth, and recognition networks concurrently on a small device. This is attractive for a smart bin that needs multimodal inference without a GPU.

- **Standard Raspberry Pi (CPU) or Arduino:** Simpler systems might offload heavy processing and just use basic sensors. For example, an Arduino can read an IR sensor or load cell to decide if bin is full, without any vision. A Raspberry Pi 4 (no accelerator) can run a small CNN for classification at a lower framerate (perhaps 1-2 FPS for ResNet-18). If using such limited hardware, the model choices are limited to very small CNNs (Tiny-YOLO, SqueezeNet, etc.) or one can leverage cloud for inference (with the drawbacks mentioned).

Edge Software optimization: When deploying on edge, optimizations like **quantization, pruning, and efficient libraries** are critical. Quantizing models to INT8 can dramatically speed up inference on CPUs and some NPUs, at minor accuracy cost. Pruning (removing redundant filters) can lighten the model. Frameworks like **TensorRT**, **OpenVINO** (for Intel hardware), and **ONNX Runtime** provide accelerated inference on various platforms. For instance, one could convert a PyTorch waste classifier to ONNX, then use TensorRT on Jetson to run it 2x faster than the raw PyTorch model. Additionally, using **accelerator-specific APIs** (e.g., NVDLA engines on Jetson or the Edge TPU API for Coral) is often needed to fully utilize the hardware.

Real-time and concurrency: An edge system often needs to perform multiple tasks: e.g., detect an object, segment it, estimate depth, and read sensors – all ideally in under a second as an item is dropped. Careful pipeline design is needed: using **asynchronous threads or pipelines** (one thread grabs images and runs detection, another runs depth on the previous frame, etc.) can maximize throughput. For example, a pipeline could use the camera's video feed to continuously estimate depth in the background, but only run the heavier detection network when a sensor triggers (item detected entering). This kind of design ensures important events are processed without unnecessary constant computation.

Power and thermal considerations: Smart bins may be battery-powered or energy-scavenging (solar). Devices like Jetson Orin may draw 10–15 W when running heavy AI loads, which might be too high for off-grid usage without a good battery. In contrast, a Raspberry Pi or Coral runs around 5 W or less for moderate tasks. Duty cycling (sleeping when idle) and using low-power modes (Jetson can be set to 5 W mode, limiting clocks) can help. Thermal management is also critical; enclosed bins can get warm, so ensuring the processor has proper cooling (heat sinks, maybe ventilation) will prevent throttling.

Ruggedness and maintenance: In a waste environment, sensors might get dirty or obstructed. Cameras may need protective covers; depth sensors like RealSense have occlusion issues if mud or debris covers the projector or lens. Implementation insights from real products like CleanRobotics' TrashBot indicate the need for self-cleaning or easy maintenance design [100] [101]. The algorithms may also need to handle partial data (e.g., if the depth camera fails, fall back to a simpler method).

To summarize, deploying these AI models on the edge requires choosing the right hardware (balancing **compute vs cost vs power**), optimizing models for that hardware, and designing a robust pipeline that meets **real-time requirements**. Fortunately, the trends in hardware (more efficient NPUs) and software (better model compression) are making it increasingly feasible to run fairly advanced multi-modal inference (vision, audio, etc.) on a compact device like a smart bin controller.

Benchmarking and Evaluation

Assessing the performance of waste characterization systems requires looking at multiple aspects:

- **Classification metrics:** For waste classification tasks (e.g., predicting the correct category of an item), standard metrics include **accuracy, precision, recall, and F1-score** for each class. Given class imbalance (some waste types are rarer), the use of macro-averaged F1 or per-class recall is important to ensure even rare hazardous items are detected. In some studies, overall accuracy is reported (often in the high 90s for small datasets), but in realistic tests with more classes, confusion matrices are examined to see, for example, if plastic vs paper are being confused.

- **Detection and segmentation metrics:** When evaluating object detectors on datasets like TACO, the common metric is **mean Average Precision (mAP)** at certain IoU thresholds (e.g. mAP@50 or the COCO-style mAP@[50:95]). TACO's authors, for instance, used mAP to compare models and noted how instance segmentation can improve detection of heavily overlapping trash [22]. For segmentation, **mean Intersection-over-Union (mIoU)** or the average precision on masks (which is analogous to detection AP but for the mask overlap) is used. A challenge in waste segmentation evaluation is defining classes: since some datasets have a very fine taxonomy (60 classes in TACO), researchers often evaluate at the **"super-category" level (e.g. plastic, paper, metal)** to get more meaningful performance numbers [102] [103]. This also mitigates the impact of misclassifying a specific subtype (e.g., mistaking a "plastic lid" for a "plastic bottle" is less severe if both fall under "plastic").

- **Depth and volume metrics:** For depth estimation models, standard metrics include **absolute error (in meters), relative error, and scale-invariant RMSE**. In the context of volume estimation, error is often given as a percentage of the true volume. Kim et al. (2025) reported a 2.37% average volume error for their RGB-D method [104] [105], which is excellent. Ultrasonic fill level sensors might be evaluated by how many centimeters error in fill height or how stable the reading is. If using monocular depth, one might evaluate the **depth maps qualitatively** or compare key distance measurements to ground truth (e.g., measure the height of an object in the image via depth vs a known measurement). In waste facility applications, an important benchmark is the **consistency of volume estimates** – for example, a case study found significant discrepancies between volume measured in a bunker vs after baling, pointing out the need for standardized calibration [106] [107]. Therefore, evaluating volume estimation might involve cross-checking with weight (if density known) or with another method (laser scan vs ultrasonic).

- **Weight estimation metrics:** Since weight prediction is a regression, metrics like **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **Mean Absolute Percentage Error (MAPE)** are used. Additionally, **R² (coefficient of determination)** is reported to show how well the variation in weight is explained by the model. In Wimalasiri's food weight estimation, they achieved RMSE ~6.32 (grams) and MAPE ~0.064% [108], and R² ≈ 0.9865, indicating a very tight prediction [109]. For scrap metal, Diaz-Romero et al. got R² ~0.71 [110], reflecting the harder variability in shape and composition. It's important to test weight models on items *outside the training set* to ensure they generalize – e.g., using one set of objects to train and a disjoint set to test. Also, if weight estimation is integrated in a classification pipeline, errors can propagate; thus an evaluation might consider scenarios like "if classification is wrong, how badly does that affect weight estimate?".

- **System-level evaluation:** Ultimately, for a deployed system, one can evaluate **sorting accuracy** or **throughput**. For instance, a smart bin could be tested over a week: how many items were sorted correctly (into correct internal bin) vs incorrectly. CleanRobotics reported their TrashBot achieved **95% sorting accuracy** in live environments after iterative improvements [111]. Another metric is speed – how quickly can the system process each item (if too slow, items might queue up). **Latency per item** is measured from the moment an item is deposited to the system decision; many aim for under 1–2 seconds. **Energy consumption** can also be a metric if evaluating an edge solution – papers have started to measure **energy per inference** on boards like Jetson (e.g., how many mWh per image for a given model) [99]. This is relevant for battery-powered deployments.

- **Robustness tests:** Evaluation protocols may include varying the conditions: different lighting (day/night if outdoors), presence of dirt or liquids on items, multiple items at once (for systems that handle that), etc. A truly robust waste characterization system should be stress-tested on these scenarios. While not standardized, some research reports include tests on **real-world**

noisy data – e.g., Lapo et al. (2024) found that a model with 97% lab accuracy dropped to 80% in real-world due to background diversity [112] [113]. Including such tests in evaluation is crucial for realistic performance assessment.

In academic challenges and benchmarks, one might see a *leaderboard* for litter detection (e.g., the TACO challenge if it existed could rank by mAP). For weight estimation, since no large public benchmark exists, comparison is usually against prior works on similar tasks (like fish weight or food weight). Each sub-problem (classification, detection, depth, weight) might be evaluated separately, but an integrated system needs a holistic evaluation, ensuring **the end-to-end pipeline meets the application's requirements (accuracy and speed)**.

Conclusion

Developing a real-world waste characterization and mass estimation system requires **combining multiple sensing and AI techniques** into a robust pipeline. From the survey above, it is clear that: (1) **Vision-based recognition** of waste is mature enough to achieve high accuracy using deep models, especially when leveraging large datasets or pre-trained vision-language models for generalization. (2) **Depth sensing or estimation** is a pivotal component for measuring physical properties – affordable options like monocular depth networks or stereo cameras can approximate volume, while RGB-D sensors provide more precise data for volume calculation and even enable detecting the difference between a full vs empty container. (3) **Mass estimation** can be approached through learned models, but incorporating physical reasoning (volume, density) or simply measuring weight directly greatly improves reliability. (4) **Sensor fusion** – be it adding a weight scale, acoustic sensor, or IR – enhances the system's ability to handle the diverse and messy nature of waste (for example, audio can clue in on glass vs plastic, weight sensors remove ambiguity in mass).

On the implementation side, deploying these capabilities at the edge is increasingly feasible with modern compact AI hardware. We must pay attention to **model efficiency, real-time processing, and environmental durability**. Edge AI systems like smart bins should be designed with modular components (so that, say, if depth estimation fails, the system can fall back on simpler logic) and fail-safes (to avoid mis-sorting hazardous waste).

Future directions in this field include: using **reinforcement learning with robotic arms** to actively interact with waste (weighing or squeezing items to deduce material), applying **continual learning** so the system improves as it sees new types of waste, and integrating the data into broader **IoT waste management platforms** (providing real-time waste composition and weight statistics for better recycling logistics). By combining the advances in computer vision, multimodal sensors, and edge computing, the next generation of smart waste systems will be more accurate, adaptable, and efficient – moving us closer to automated recycling and improved sustainability in waste management [114] [115].

References: The information and data points in this report are drawn from a range of recent studies and open-source resources in the field, including academic journals and conference papers in 2020–2025, as well as relevant code repositories and project reports. Key sources include the TACO dataset paper [23], multi-model detection frameworks for waste [17], vision-language model evaluations for waste sorting [35], depth-based volume estimation research [43] [42], and integrated scrap classification and weighing studies [75] [77], among others cited throughout the text. These provide a foundation of peer-reviewed evidence for the methods and performance metrics discussed.

---

[1] [2] [3] [16] [17] [18] [19] [20] [84] [85] Automatic Detection and Classification System of Domestic Waste via Multimodel Cascaded Convolutional Neural Network
https://web.comp.polyu.edu.hk/pinli/CoRR/TII/TII2022_1.pdf

[4] [8] [9] [14] [55] [56] [57] [58] [59] [60] [61] [66] [67] [68] [69] [70] [71] [72] [73] [74] [75] [76] [77] [78] [79] [80] [110] upcommons.upc.edu
https://upcommons.upc.edu/bitstreams/d2042efa-4c93-4788-bd22-fddd3bc6c8a3/download

[5] [6] [7] [10] [11] [12] [15] [26] [28] [112] [113] 10.54287/gujsa.1721478
https://dergipark.org.tr/en/download/article-file/4967216

[13] [21] [22] [23] GitHub - pedropro/TACO: Trash Annotations in Context Dataset Toolkit
https://github.com/pedropro/TACO

[24] [93] [94] [95] Top Waste Datasets and Models | Roboflow Universe
https://universe.roboflow.com/search?q=class%3Awaste

[25] [PDF] Feature Design for Bridging SAM and CLIP toward Referring Image ...
https://openaccess.thecvf.com/content/WACV2025/papers/Ito_Feature_Design_for_Bridging_SAM_and_CLIP_toward_Referring_Image_WACV_2025_paper.pdf

[27] Integrating artificial intelligence for sustainable waste management
https://www.sciencedirect.com/science/article/pii/S2589471425000270

[29] [30] [31] [32] [33] [34] [35] [36] [37] [38] Enhancing waste recognition with vision-language models: A prompt engineering approach for a scalable solution
https://researchers.cdu.edu.au/files/129250152/1-s2.0-S0956053X25003502-main.pdf

[39] [40] [62] [63] [64] [65] [108] [109] Vision-Based Approach for Food Weight Estimation from 2D Images Identify applicable funding agency here. If none, delete this.
https://arxiv.org/html/2405.16478v1

[41] [42] [43] [44] [45] [104] [105] Volume Estimation Method for Irregular Object Using RGB-D Deep Learning
https://www.mdpi.com/2079-9292/14/5/919

[46] Volume measurement of soil in excavator buckets by solid-state 3D ...
https://link.springer.com/article/10.1007/s41693-025-00166-w

[47] [48] [91] [92] [99] [102] [103] [114] [115] Estimer poids et matière (1).pdf
file://file_00000000879061f5a9720c93b63d44d5

[49] [50] [51] [52] [53] [54] [106] [107] Volume Determination Challenges in Waste Sorting Facilities: Observations and Strategies - PMC
https://pmc.ncbi.nlm.nih.gov/articles/PMC11014339/

[81] Innovation of the Day | CleanRobotics - TrendWatching
https://www.trendwatching.com/innovation-of-the-day-cleanrobotics

[82] The Future of Waste Sorting on Campus: AI Smart Bins
https://wasteadvantagemag.com/the-future-of-waste-sorting-on-campus-ai-smart-bins/

[83] Tongs for Recognizing Littering Garbage with Active Acoustic Sensing
https://dl.acm.org/doi/10.1145/3567445.3571112

[86] [87] WasteGAN: Data Augmentation for Robotic Waste Sorting ... - arXiv
https://arxiv.org/html/2409.16999v1

[88] Data Augmentation Using Background Replacement for Automated ...
https://www.mdpi.com/2313-433X/7/8/144

89 A comprehensive approach for waste management with GAN ...
https://pmc.ncbi.nlm.nih.gov/articles/PMC12453751/

90 Adding Material Embedding to the image2mass Problem | Request ...
https://www.researchgate.net/publication/362193880_Adding_Material_Embedding_to_the_image2mass_Problem

96 MiDaS - PyTorch
https://pytorch.org/hub/intelisl_midas_v2/

97 98 DepthAnything/Depth-Anything-V2 - GitHub
https://github.com/DepthAnything/Depth-Anything-V2

100 CleanRobotics sweeps up the AI garbage game with two new ...
https://technical.ly/uncategorized/cleanrobotics-sweeps-up-the-ai-garbage-game-with-two-new-trashbot-models/

101 CleanRobotics TrashBot Smart Recycling Bin | Semtech
https://www.sierrawireless.com/resources/customer-stories/cleanrobotics-trashbot/

111 CleanRobotics Adds Educational Feature to Trashbot Robotic ...
https://www.robotics247.com/article/cleanrobotics_adds_educational_feature_to_robotic_recyling_bin